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Smartphone addiction as a predictor of anxiety, depression, and sleep disorders: evidence from a cross-sectional study among Egyptian university students

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Abstract

Background With the widespread use of smartphones, concerns have emerged regarding their potential impact on mental health and sleep. This study examined the associations between smartphone addiction and symptoms of anxiety, depression, and sleep disorders among university students.

Methods A descriptive correlational design was employed with a convenience sample of 460 Egyptian university students from Fayoum University & Ain Shams University. The total sample consisted of (140 males, 320 females) aged 19–24 years. Data were collected using validated Arabic versions of the Smartphone Addiction Scale, GAD-7, the depression subscale of the HSCL-25, and a Sleep Disorders Scale. Pearson correlations and simple linear regression analyses were conducted.

Results Smartphone addiction was positively associated with anxiety ($r = 0.485$), depression ($r = 0.627$), and sleep disorders ($r = 0.567$) ($p < 0.001$). Regression analyses indicated that smartphone addiction accounted for 26%, 23% & 30% of the variance of anxiety, depression, and sleep disorders. These results indicate that smartphone addiction may serve as a potential risk marker for anxiety, depression, and sleep disorders.

Conclusion Higher levels of smartphone addiction were significantly associated with increased psychological distress and sleep disturbances. While causal inferences cannot be drawn due to a cross-sectional design, smartphone addiction may function as a potential risk marker within a broader network of psychological vulnerability. These findings highlight the need for longitudinal and multivariate research to better clarify the underlying mechanisms and directional relationships.

Keywords Smartphone Addiction, Addictive Behavior, Anxiety Disorders, Depressive Disorder, Sleep disorders, University Students, Epidemiology

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Introduction

Smartphone technology now occupies a central position in everyday life, especially for university students. Despite the clear academic, social, and practical advantages these devices offer, research indicates that excessive and poorly regulated use may be associated with negative psychological consequences.

In this context, problematic smartphone use has been increasingly understood as part of the broader spectrum of behavioral addictions, reflecting patterns of compulsive involvement, diminished self-regulation, tolerance-like processes, and interference with daily functioning.

Given the efficiency with which smartphones perform these tasks in daily living, they have achieved remarkably widespread popularity worldwide, with the number of users continuing to increase day by day [1].

Ayandele et al. [2] indicated that the average daily frequency of smartphone use reaches approximately 150 times per day. A substantial proportion of users spend nearly six hours or more browsing smartphone applications, to the extent that they find it difficult to dispense with them.

Smartphone use has expanded rapidly worldwide over the past decade, with adoption continuing to grow across both developed and developing regions. Contemporary estimates suggest that approximately 70% of the global population—around 5.78 billion individuals—were using smartphones by 2026, compared to 3.8 billion (49%) in 2020, reflecting a substantial increase in a relatively short period [3]. This growth is further reflected in the total number of devices, which reached approximately 7.58 billion, indicating that many individuals own multiple devices for personal and professional use.

Despite variations across regions, smartphone penetration has risen markedly in developing countries, with earlier projections suggesting rates approaching 90% in some contexts. Behavioral patterns also indicate intensive daily engagement, as users in Europe interact with their smartphones more than 48 times per day, and approximately 65% report checking their devices immediately upon waking or within the first 15 min thereafter [4]. These trends suggest that smartphone use has become a pervasive aspect of daily life.

Smartphones constitute a double-edged sword; while they facilitate many aspects of daily life, they also give rise to a range of problems and disturbances that are cause for concern and psychological distress because of excessive use and the development of smartphone addiction [5].

As a result, the sharp rise in smartphone use in daily life, together with the rapid expansion of their applications, has prompted an increasing volume of research focused on the potential detrimental effects of smartphones on psychological well-being and social functioning.

The smartphone has effectively become both a telephone and a computer simultaneously, a convergence that has further reinforced patterns of smartphone addiction, which has increasingly come to represent a significant social threat [6].

Smartphone addiction is a type of behavioral addiction that involves excessive and compulsive use of smartphones, resulting in negative psychological, social, emotional, cognitive, and health-related effects [7].

Although problematic smartphone use shares several features with behavioral addictions, it has not been formally recognized as an independent diagnostic category in the ICD-11. While the World Health Organization has officially classified Gaming Disorder under "Disorders due to addictive behaviors," problematic smartphone use continues to be understood within the broader conceptual framework of behavioral or technology-related addictions rather than as a standalone diagnosis.

This lack of formal diagnostic recognition has important implications for research and interpretation. Specifically, it suggests that smartphone addiction should be conceptualized as a dimensional and behaviorally defined construct rather than a discrete clinical disorder. Accordingly, findings in this area should be interpreted with caution, avoiding direct clinical generalizations or diagnostic labeling.

Therefore, the term "smartphone addiction" in the current study is used to describe a pattern of excessive and compulsive use associated with functional impairment rather than a formally recognized psychiatric diagnosis [8].

Recent studies have shown that university students are most vulnerable to smartphone addiction, as they spend substantial amounts of time interacting with smartphones, a pattern accompanied by a range of negative consequences resulting from excessive use. Such overuse has been found to adversely affect academic performance, physical health, psychological well-being, and social functioning, thereby potentially hindering their engagement in daily life activities [7].

Epidemiological statistics on the prevalence of this phenomenon among university students indicate that approximately 80% of students use smartphones for periods ranging from five to seven hours per day, meaning that nearly one-third of the day is spent using a smartphone [9].

Elwakeel et al. [7] demonstrated that 36.8% of some students from Jeddah University used smartphones between one and seven hours daily, whereas 63.2% of the sample reported using smartphones for more than seven hours per day; these findings indicated markedly high levels of smartphone use among Jeddah University students.

The findings of numerous clinical studies demonstrated that smartphone addiction is associated with a range of psychological disorders. Anxiety, depression, impulsiveness, and sleep disorders have been identified as the most prevalent psychological conditions linked to smartphone addiction [10–12].

Several theoretical perspectives have been advanced to clarify the psychological mechanisms that may underline excessive smartphone use. The Compensatory Internet Use Model, for instance, suggests that individuals may turn to intensive technology use as a means of alleviating negative emotions or coping with psychological discomfort. In a similar vein, cognitive–behavioral approaches emphasize the role of reinforcement processes, fear of missing out (FOMO), and maladaptive coping styles in fostering persistent and compulsive patterns of use. In relation to sleep outcomes, the Pre-Sleep Cognitive Arousal Model provides further insight by proposing that smartphone engagement during evening hours may interfere with sleep through heightened cognitive stimulation and physiological activation [7].

Empirical research has repeatedly shown that smartphone addiction is significantly linked to elevated symptoms of anxiety, depression, and sleep disorders among university students across diverse cultural contexts [6, 12–25].

Nevertheless, most of these studies have focused on populations in Western and East Asian countries, leaving evidence from Arab cultural settings relatively scarce. This gap is notable given the region's rapid digital growth and high rates of smartphone adoption. Moreover, cultural factors may potentially influence patterns of smartphone use and associated psychological responses.

Within this context, an important question emerges regarding the extent to which excessive smartphone use is associated with psychological and behavioral outcomes. These influences were not directly examined in the present study.

Considering these contextual factors, investigating smartphone addiction among Arab university students holds considerable theoretical and clinical significance. Gaining insight into these relationships can aid in the early identification of psychological risk markers and support the development of culturally tailored preventive and intervention strategies.

Accordingly, the present study seeks to investigate the associations between smartphone addiction and symptoms of anxiety, depression, and sleep disturbances among Egyptian university students, as well as to assess the degree to which smartphone addiction accounts for variability in these psychological outcomes.

While these theoretical models originate from different perspectives, they share common underlying mechanisms, including emotional regulation, cognitive overactivation, and maladaptive behavioral reinforcement. In the present study, these frameworks are considered complementary rather than competing, providing an integrated lens through which the associations between smartphone addiction and psychological outcomes can be interpreted.

Based on the reviewed literature, an integrative conceptual model is proposed to guide the interpretation of the relationships examined in the present study (see Fig. 1).

Research aim

This study aims to examine the statistical associations between smartphone addiction and symptoms of anxiety, depression, and sleep disorders among Egyptian university students, and to assess the extent to which smartphone addiction contributes to the variance observed in these psychological outcomes.

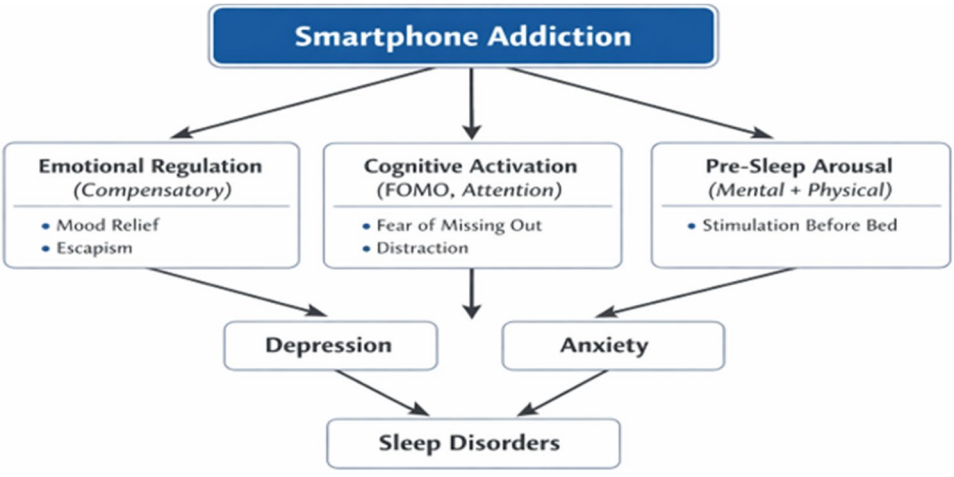


Fig. 1 The conceptual model illustrates mechanisms linking smartphone addiction to anxiety, depression, and sleep disturbances. The model integrates compensatory emotional regulation, cognitive activation processes, and pre-sleep arousal as complementary explanatory pathways

Significance

The significance of the present study lies in its contribution to both theory and practice. From a theoretical perspective, it extends the existing literature by examining the interplay between smartphone addiction, psychological distress, and sleep disturbances within an underrepresented Arab cultural context. From a practical perspective, the findings may inform early identification efforts and guide the development of preventive and supportive interventions targeting problematic smartphone use and its associated psychological outcomes among university students.

Research hypotheses

1 -Smartphone addiction is positively associated with anxiety, depression, and sleep disorders.
2 -Smartphone addiction significantly predicts levels of anxiety, depression, and sleep disorders

Method

Research design

The current study utilized a descriptive approach with a correlational design, deemed suitable for addressing the research objectives and for effectively examining the proposed research questions and hypotheses.

Sample

A convenient sample of university students was recruited for the study, comprising a total of 460 participants from Fayoum University and Ain Shams University. The sample included 140 males and 320 females, with ages ranging from 19 to 24 years (M = 20.17, SD = 1.62). The demographic characteristics of the participants are summarized in Table 1.

Table 1 Demographic characteristics of the study sample (*N* = 460)

Variable	Category	Frequency (f)	Percentage (%)
Age	M = 20.17 years SD = 1.162 years		
Gender	Male Students	140	33.3%
	Female Students	320	66.7%
University	Fayoum	340	73%
	Ain Shams	120	26%
Major	Psychology	170	36.9%
	Sociology	75	16.3%
	Medicine	40	8.7%
	Nursing	120	26.1%
	Pharmacy	55	11.9%
Academic Year	First Year	170	36.9%
	second Year	105	22.9%
	Third Year	115	25%
	Fourth year	70	15.2%
Marital Status	Single	429	93.3%
	Married	31	6.7%
Residence	Rural	297	64.6%
	Urban	163	35.4%

Table 1 presents a detailed demographic profile of the study sample, highlighting key characteristics such as age, gender, university affiliation, field of study, academic level, marital status, and place of residence. This information offers a clear depiction of the sample, showing relative homogeneity in terms of age, gender, and marital status, alongside variation in academic disciplines and residential backgrounds, factors that may influence psychological and behavioral outcomes. While a convenience sampling approach was used, participants were drawn from multiple academic fields to increase heterogeneity. However, the uneven gender distribution represents a limitation for generalizability, and potential gender-related effects were examined analytically to mitigate bias.

The use of convenience sampling was primarily driven by practical and logistical considerations, including accessibility to participants and institutional constraints. To partially mitigate the limitations associated with this approach, the sample was drawn from two universities and included students from diverse academic disciplines, thereby enhancing variability within the sample. Nevertheless, this sampling method may limit the representativeness of the findings and should be considered when interpreting the results.

The observed gender imbalance, with a higher proportion of female participants, represents a potential source of bias. To address this issue, supplementary analyses were conducted to examine whether gender differences significantly influenced the main study variables. The results indicated no substantial gender-based differences in smartphone addiction, anxiety, depression, or sleep disorders. However, despite these findings, the unequal distribution remains a limitation that may affect the generalizability of the results. Future research is encouraged to employ more balanced sampling strategies to ensure greater representativeness across genders.

Statistical analysis plan

Consistent with the study hypotheses, Pearson correlation coefficients were computed to assess the relationships between smartphone addiction and symptoms of anxiety, depression, and sleep disturbances. To evaluate the predictive role of smartphone addiction, simple linear regression analyses were performed, treating smartphone addiction as the independent variable and each psychological outcome as a separate dependent variable. Before conducting the regression analyses, all relevant assumptions, including normality, linearity, homoscedasticity, and the absence of multicollinearity, were thoroughly examined and satisfied.

The use of simple linear regression models was intended to provide an initial examination of the direct statistical associations between smartphone addiction and each psychological outcome. While this approach allows for a clear and parsimonious assessment of bivariate relationships, it does not account for the potential influence of confounding variables.

No statistical weighting or adjustment procedures were applied to correct for sampling imbalance, which should be considered when interpreting the findings.

In addition to examining normality, linearity, and homoscedasticity, residual distributions were visually inspected to ensure no major deviations from model assumptions. However, confidence intervals for regression estimates and additional robustness checks were not included, which should be considered when interpreting the stability of the findings.

In addition to the primary analyses, diagnostic procedures were conducted to further evaluate the robustness and stability of the regression models. Residual distributions were visually inspected using histograms and normal probability plots, which indicated no substantial deviations from normality. Furthermore, scatterplots of standardized residuals versus predicted values were examined to assess homoscedasticity, with no serious violations detected.

To enhance the interpretability of the findings, confidence intervals were also considered as an important complement to statistical significance testing, providing a range of plausible values for the estimated effects.

The use of simple linear regression in the present study was guided by the primary objective of examining the direct statistical association between smartphone addiction and each psychological outcome independently. This approach allows for a clear and interpretable estimation of the proportion of variance explained by smartphone addiction without the influence of additional covariates. While more advanced techniques such as multiple regression or structural equation modeling (SEM) may provide a more comprehensive analysis by accounting for confounding variables and complex interrelationships, these methods were beyond the scope of the current study. Future research is encouraged to employ multivariate and model-based approaches to further explore these relationships in greater depth.

In addition to significance testing, 95% confidence intervals (CIs) were estimated for all regression coefficients to improve the precision and interpretability of the observed effects. Standard errors (SEs) were also reported for both unstandardized and standardized coefficients where applicable. Although each model included a single predictor, collinearity diagnostics were additionally examined to confirm the absence of multicollinearity concerns.

Instruments

Demographic characteristics assessment questionnaire

The questionnaire comprised items designed to capture the demographic characteristics of the participants, including age, gender, place of residence, university affiliation, and academic major. This instrument does not yield a total or composite score.

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Smartphone Addiction Scale (SAS): by Kwon et al., [26]

The Smartphone Addiction Scale (SAS) is a self-report instrument developed to evaluate the severity of smartphone addiction. It consists of 33 items across six distinct dimensions, with responses provided on a six-point Likert scale ranging from 1 ("Strongly Disagree") to 6 ("Strongly Agree"), resulting in a total score range of 33 to 198. The six dimensions are: Daily Life Disturbance (5 items), Positive Expectancy (8 items), Withdrawal (6 items), Cyberspace-Oriented Relationship (7 items), Overuse (4 items), and Tolerance (3 items).

The Smartphone Addiction Scale (SAS) is a self-report instrument developed to evaluate the severity of smartphone addiction. It consists of 33 items across six distinct dimensions, with responses provided on a six-point Likert scale ranging from 1 ("Strongly Disagree") to 6 ("Strongly Agree"), resulting in a total score range of 33 to 198. The six dimensions are: Daily Life Disturbance (5 items), Positive Expectancy (8 items), Withdrawal (6 items), Cyberspace-Oriented Relationship (7 items), Overuse (4 items), and Tolerance (3 items).

Kwon et al. [26] reported excellent internal consistency for the total scale (Cronbach's $\alpha = 0.967$) and strong reliability for each subscale ($\alpha = 0.858$, 0.913, 0.904, 0.876, 0.825, 0.885, respectively). The scale's concurrent validity was supported through significant correlations with both the Smartphone Addiction Proneness Scale (SAPS) and the Korean Self-Reporting Internet Addiction Scale, with all correlations reaching statistical significance at $p < 0.01$, confirming the instrument's robustness and validity.

Elwakeel [9]. Verified the psychometric properties of the Arabic version of the SAS within an Egyptian context. Construct validity was assessed using confirmatory factor analysis (CFA), which supported the hypothesized six-factor structure and demonstrated an excellent fit to the data. Fit indices indicated strong model adequacy. Regarding reliability, the overall Cronbach's alpha for the scale was 0.939, while the split-half reliability coefficient was 0.895, demonstrating excellent internal consistency and confirming the scale's suitability for research purposes.

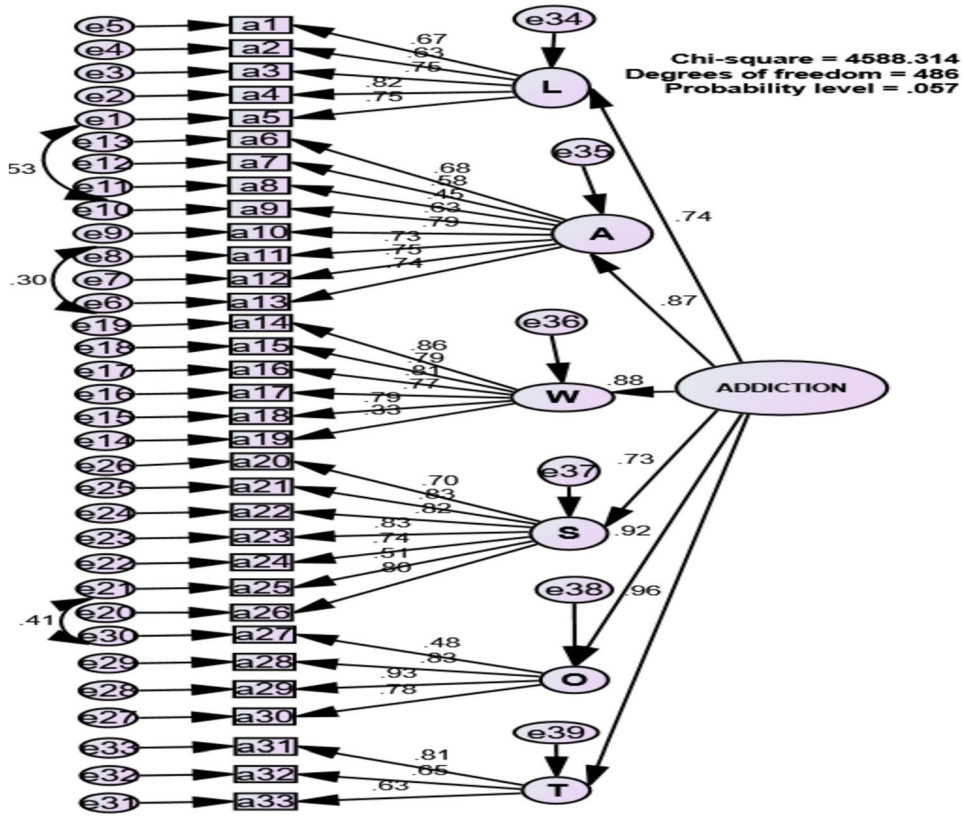


Fig. 2 Hypothesized model of the depression scale using Amos 24 software

Table 2 Indicators of the smartphone addiction scale model fit among the sample

X^2/df	GFI	AGFI	NFI	CFI	IFI	TLI	RMSEA
2.44	0.965	0.961	0.958	0.969	0.969	0.966	0.049

Validity in the present study

The confirmatory factor analysis (CFA) was conducted on the sample before starting the analysis. The researchers designed the smartphone addiction scale model according to the theoretical framework of the scale; we can see that from the following figure: (Fig. 2).

The confirmatory factor analysis was conducted among the sample through AMOS 24, and it showed good model fit indicators and achieved the confirmatory factor validity, as shown in the following table: (Table 2)

The previous table shows that the proposed model of the smartphone addiction scale demonstrates strong evidence of good fit within the sample. The Chi-square value was not statistically significant, and the ratio of Chi-square to degrees of freedom was 2.44, which is below the commonly accepted threshold of 5, supporting the adequacy of the model.

In addition, several fit indices including the Goodness of Fit Index (GFI), Adjusted Goodness of Fit Index (AGFI), Normed Fit Index (NFI), Comparative Fit Index (CFI), Incremental Fit Index (IFI), and Tucker–Lewis Index (TLI) all yielded high values approaching 1, indicating a strong alignment between the model and the observed data.

Moreover, the Root Mean Square Error of Approximation (RMSEA), a key indicator in confirmatory factor analysis, was 0.049. This low value reflects an excellent level of fitness, suggesting that the model closely represents the data. Overall, these findings provide solid support for the structural validity of the smartphone addiction scale within the study sample.

Reliability

In the present study, the reliability of the SAS was evaluated using Cronbach's alpha across the entire sample ($N = 460$), resulting in a high coefficient of 0.887. This demonstrates strong internal consistency, confirming the scale's appropriateness for use with the study participants and supporting the reliability of the obtained data.

Generalized anxiety scale (GAD-7): developed by Spitzer et al. [27]

The Generalized Anxiety Disorder Scale (GAD-7) is composed of 7 items designed to assess symptoms of generalized anxiety. Participants respond by indicating how frequently each statement applies to them, using a four-point scale: "Not at all," "Several days," "More than

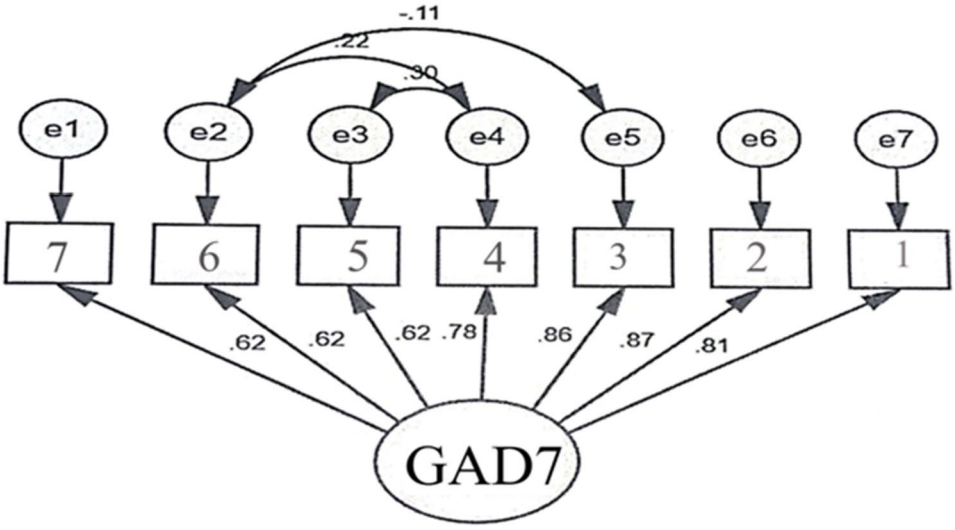


Fig. 3 Hypothesized model of the GAD-7 scale using Amos 24 software

Table 3 indicators of the GAD-7 scale model fit among the sample

X^2/df	GFI	AGFI	NFI	CFI	IFI	TLI	RMSEA
1.98	0.977	0.965	0.961	0.970	0.970	0.964	0.041

half the days," and "Nearly every day." The scale's psychometric properties have been well established in both international and Arabic contexts, with reported Cronbach's alpha values of 0.918, reflecting excellent internal consistency [28]. Elwakeel [9]. The psychometric properties of the GAD-7 were obtained within an Egyptian context. Construct validity was examined using confirmatory factor analysis (CFA), which indicated an excellent fit between the hypothesized model and the data.

Validity in the present study

The confirmatory factor analysis (CFA) was conducted on the sample before starting the analysis. The researchers designed the GAD-7 scale model according to the theoretical framework of the scale; we can see that from the following figure: (Fig. 3)

The confirmatory factor analysis was conducted among the sample through AMOS 24, and it showed good model fit indicators and achieved the confirmatory factor validity, as shown in the following table: (Table 3)

The previous table shows that the proposed model of the GAD7 scale demonstrates strong evidence of good fit within the sample. The Chi-square value was not statistically significant, and the ratio of Chi-square to degrees of freedom was 1.98, which is below the commonly accepted threshold of 5, supporting the adequacy of the model. In addition, several fit indices including the Goodness of Fit Index (GFI), Adjusted Goodness of Fit Index (AGFI), Normed Fit Index (NFI), Comparative Fit Index (CFI), Incremental Fit Index (IFI), and Tucker–Lewis Index (TLI) all yielded high values approaching 1, indicating a strong alignment between the model and the observed data. Moreover, the Root Mean Square Error of Approximation (RMSEA), a key indicator in confirmatory factor analysis, was 0.041. This low value reflects an excellent level of fitness, suggesting that the model closely represents the data. Overall, these findings provide solid support for the structural validity of the GAD7 scale within the study sample.

Relatability

In the present study, the reliability of the GAD-7 was evaluated using Cronbach's alpha across the full sample ($N = 460$), yielding a high coefficient of 0.897. This reflects strong internal consistency and confirms the scale's appropriateness for use with the study participants, ensuring the reliability of the obtained data.

Depression scale from the Hopkins Symptom Checklist-25 (HSCL-25): In: [29]

The Hopkins Symptom Checklist-25 (HSCL-25) is a self-report instrument developed to assess symptoms of anxiety and depression, comprising 25 items in total, 9 measuring anxiety and 16 assessing depression. The present study focused solely on depression subscale. Originating from the Cornell Medical Index at Johns Hopkins University, Maryland, USA, the depression subscale evaluates core depressive symptoms such as sadness, discouragement, loss of motivation, low energy, lethargy, negative thoughts, guilt, and general dissatisfaction. Items are rated on a four-point Likert scale ranging from 1 ("Not at all") to 4 ("Extremely"), producing a total possible score between 16 and 64.

Elwakeel [9]. examined the psychometric properties of the Arabic version of the HSCL-25 in the Egyptian

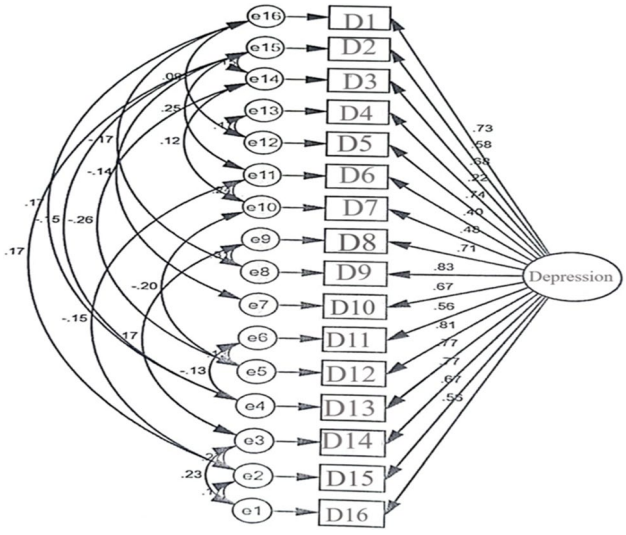


Fig. 4 Hypothesized model of the depression scale using Amos 24 software

Table 4 Indicators of the depression model fit among the sample

X^2/df	GFI	AGFI	NFI	CFI	IFI	TLI	RMSEA
3.12	0.967	0.960	0.951	0.966	0.966	0.964	0.033

context. Confirmatory factor analysis (CFA) indicated an excellent fit for the depression subscale, with a non-significant χ^2 p-value of 0.073 and high values across multiple fit indices, including GFI, AGFI, NFI, CFI, IFI, and TLI, all approaching the maximum of 1. The RMSEA was 0.029, further confirming the strong model-data fit and supporting the construct validity of the depression scale within the current study sample.

Reliability

Reliability of the depression subscale was evaluated using Cronbach's alpha and split-half methods, with the latter adjusted via the Spearman-Brown formula. The total scale demonstrated a Cronbach's alpha of 0.932, while the split-half reliability coefficient was 0.895, indicating excellent internal consistency and confirming the instrument's suitability and robustness for research applications.

Validity in the present study

The confirmatory factor analysis (CFA) was conducted on the sample before starting the analysis. The researchers designed the depression scale model according to the theoretical framework of the scale; we can see that from the following figure: (Fig. 4)

The confirmatory factor analysis was conducted among the sample through AMOS 24, and it showed good model fit indicators and achieved the confirmatory factor validity, as shown in the following table: (Table 4)

The previous table shows that the proposed model of the depression scale demonstrates strong evidence of good fit within the sample. The Chi-square value was not statistically significant, and the ratio of Chi-square to degrees of freedom was 3.12, which is below the commonly accepted threshold of 5, supporting the adequacy of the model. In addition, several fit indices including the Goodness of Fit Index (GFI), Adjusted Goodness of Fit Index (AGFI), Normed Fit Index (NFI), Comparative Fit Index (CFI), Incremental Fit Index (IFI), and Tucker–Lewis Index (TLI) all yielded high values approaching 1, indicating a strong alignment between the model and the observed data. Moreover, the Root Mean Square Error of Approximation (RMSEA), a key indicator in confirmatory factor analysis, was 0.033. This low value reflects an excellent level of fitness, suggesting that the model closely represents the data. Overall, these findings provide solid support for the structural validity of the depression scale within the study sample.

Reliability in the current study

In the present study, the reliability of the depression subscale was examined using Cronbach's alpha across the full sample ($N = 460$), resulting in a high coefficient of 0.863. This reflects strong internal consistency and confirms the scale's appropriateness for use with the study participants, ensuring the reliability of the collected data.

Sleep disorders scale [30]

The Sleep Disorders Scale is composed of 28 items aimed at evaluating the overall severity of sleep-related problems. Participants respond to each item using a five-point scale: "Never," "Rarely," "Sometimes," "Often," and "Always," scored 5, 4, 3, 2, and 1, respectively, with reverse scoring applied where applicable. The total possible score on the scale ranges from 28 to 140.

Elwakeel [9] examined the psychometric properties of the Sleep Disorders Scale in the Egyptian context, assessing both reliability and validity. Construct validity was evaluated using exploratory & confirmatory factor analyses. Exploratory factor analysis revealed four clear factors that together explained 57% of the total variance, indicating strong relationships between items and supporting the scale's validity. The four factors were:

1- Physical and Psychological Symptoms: Includes physical and psychological manifestations related to sleep, accounting for the largest portion of variance (31%)
2- Difficulty Initiating Sleep: Reflects problems falling asleep or returning to sleep after nighttime awakenings, explaining 11% of the variance.

3- Positive Sleep Symptoms: Represent restorative or beneficial aspects of sleep, explaining about 9% of the variance.
4- Sleep Satisfaction: Measures satisfaction with sleep quality and type, explaining around 5% of the variance. These findings indicate that the scale has a clear factor structure, strong item loadings, and coherent inter-item relationships, providing solid evidence for its validity in the Egyptian sample.

Confirmatory factor analysis (CFA) demonstrated that the hypothesized four-factor structure of the Sleep Disorders Scale exhibited an excellent fit to the current sample. All items loaded appropriately on their respective factors, supporting the proposed factorial structure. Fit indices indicated strong model adequacy: the chi-square to degrees of freedom ratio (χ^2/df) was 1.812, and other indices-including the Goodness-of-Fit Index (GFI), Adjusted GFI (AGFI), Normed Fit Index (NFI), Comparative Fit Index (CFI), Incremental Fit Index (IFI), and Tucker-Lewis Index (TLI)-all approached 1, reflecting an optimal correspondence between the model and the data. The Root Mean Square Error of Approximation (RMSEA) was 0.053, further confirming the model's adequacy and supporting the structural validity of the scale and its subdimensions in the Egyptian sample.

Reliability analyses indicated that the Sleep Disorders Scale demonstrated high internal consistency, with a Cronbach's alpha of 0.899 for the total score and a split-half reliability coefficient of 0.806. These results confirm the scale's stability and support its suitability for both research and clinical assessment purposes.

In the present study, the reliability of the Sleep Disorders Scale was evaluated using Cronbach's alpha across the full sample ($N = 460$), yielding a high coefficient of 0.873. This demonstrates strong internal consistency and confirms the scale's appropriateness for use with the study participants, ensuring the reliability of the collected data.

Field work

Data collection for the present study spanned five months, commencing in June 2025 and concluding at the end of November 2025. Due to logistical considerations, data were collected sequentially across the two universities. To minimize potential timing effects, identical procedures, instruments, and administration conditions were applied in both settings. Preliminary analyses revealed no significant differences between institutions in the primary study variables, supporting the comparability of the collected data.

To minimize potential response biases, participants were assured of anonymity and confidentiality, and participation was entirely voluntary. In addition, standardized and well-validated instruments were used, which may help reduce measurement error and enhance the reliability of the responses.

Table 5 Correlation between smartphone addiction, anxiety, depression, and sleep quality in the study sample ($N = 460$)

Variables	Smartphone Addiction	Anxiety	Depression	Sleep disorders
Smartphone Addiction	1	0.485***	0.627***	0.567***
Anxiety	0.485***	1	0.718***	0.543***
Depression	0.627***	0.718***	1	0.621***
Sleep disorders	0.567***	0.543***	0.621***	1

***$p < 0.001$

Results

Before performing the main analyses, key statistical assumptions were evaluated. Normality of the variable distributions was assessed using skewness and kurtosis indices, all of which fell within the acceptable range of ±2. Linearity and homoscedasticity were examined through scatterplots, confirming appropriate linear relationships among the variables. These assessments supported the suitability of using Pearson correlation coefficients and linear regression analyses for the study.

Given that all variables were assessed using self-report measures at a single time point, common method variance should be considered when interpreting the strength of the observed associations.

The first hypothesis stated that: Smartphone addiction is positively associated with anxiety, depression, and sleep disorders. To verify this hypothesis, Pearson's correlation coefficients were calculated between variables. The results are presented in Table 5.

To provide a more precise estimation of the observed associations, 95% confidence intervals were calculated for the correlation coefficients. Smartphone addiction was positively associated with anxiety ($r = 0.485$, 95% CI [0.41, 0.55]), depression ($r = 0.627$, 95% CI [0.57, 0.68]), and sleep disorders ($r = 0.567$, 95% CI [0.50, 0.63]). These intervals suggest that the observed relationships are not only statistically significant but also relatively stable within the population.

These findings highlight a closely interconnected network among smartphone addiction, mental health, and sleep quality, aligning with theoretical frameworks that suggest excessive smartphone use can adversely affect psychological well-being and sleep patterns. This evidence establishes a robust foundation for examining the predictive influence of smartphone addiction on related psychological outcomes.

The second hypothesis posited that Smartphone addiction significantly predicts levels of anxiety, depression, and sleep disorders. To test this hypothesis, simple linear

Table 6 Simple linear regression of smartphone addiction as a predictor of anxiety, depression, & sleep disorders (*N* =460)

Independent Variable	Dependent Variable	B	SE(B)	β	SE(β)	95% CI for B	R^2	F	t	*p*	VIF
Smartphone Addiction	Anxiety	0.26	0.017	0.51	0.034	[0.23, 0.29]	0.26	221.3	14.87	< 0.001	1.00
	Depression	0.23	0.017	0.48	0.035	[0.20, 0.26]	0.23	185.6	13.62	< 0.001	1.00
	Sleep Disorders	0.30	0.018	0.55	0.032	[0.27, 0.34]	0.30	286.9	16.94	< 0.001	1.00

regression analyses were conducted. The results are summarized in the Table 6:

To provide a more comprehensive interpretation of the regression findings, 95% confidence intervals were calculated for all regression coefficients. All intervals excluded zero, supporting the statistical reliability of the observed associations. Standard errors were small across models, indicating stable parameter estimation. Because each regression model included only one predictor, variance inflation factors (VIF = 1.00) confirmed the absence of multicollinearity concerns.

To provide a more precise estimation of the regression effects, confidence intervals were considered alongside the reported coefficients. The pattern of results remained consistent, indicating stable and reliable associations across the examined models. Additionally, residual diagnostics supported the adequacy of the regression models. Visual inspection of residual plots suggested that the assumptions of normality and homoscedasticity were reasonably met, with no evidence of systematic bias in the residual distribution.

Regression analyses indicated that smartphone addiction was significantly associated with levels of anxiety, depression, and sleep disorders among university students.

To examine the robustness of the findings, sensitivity analyses were performed by re-estimating the regression models separately across gender groups and university subsamples. The direction, magnitude, and statistical significance of the associations remained largely stable across subgroup analyses, indicating that the main findings were not attributable to a specific demographic subgroup.

Although confidence intervals were considered for correlation estimates, they were not consistently reported for all regression parameters, which may limit the precision of interpretation.

The observed R^2 values reflect medium to large effect sizes, demonstrating that smartphone addiction may serve as a potential risk marker for psychological distress and sleep disorders among university students.

Discussion

The present study examined the associations between smartphone addiction and symptoms of anxiety, depression, and sleep disturbances among university students, and explored the extent to which smartphone addiction was statistically related to these outcomes within an Arab cultural context. Overall, higher levels of smartphone addiction were significantly associated with greater psychological distress and poorer sleep functioning. These findings support the study hypotheses and are consistent with previous research.

The results are best understood in terms of potential underlying mechanisms rather than as isolated statistical associations. Excessive smartphone use may reflect broader difficulties in emotional regulation and behavioral control, both of which are central components of transdiagnostic models of psychological distress. From this perspective, problematic smartphone use may represent one element within a broader network of maladaptive coping responses rather than an independent phenomenon.

Consistent with prior literature, higher levels of smartphone addiction were associated with elevated anxiety and depression [31, 32]. Individuals with problematic smartphone use often report difficulties regulating behavior, which may negatively affect mental, cognitive, and physical health, as well as interpersonal relationships and academic functioning [33]. Smartphone addiction has also been linked to social anxiety among university students [34]. The convergence of these findings with earlier studies strengthens confidence in the robustness of the observed associations.

One plausible explanation is that emotionally vulnerable individuals may rely on smartphones for avoidance, reassurance seeking, or mood regulation. Excessive engagement with digital content may, in turn, intensify distress through cognitive overload, upward social comparison, disrupted concentration, and reduced participation in restorative offline activities. This pattern suggests a reciprocal process in which psychological distress and problematic smartphone use may reinforce one another over time.

The findings regarding sleep disturbances are likewise consistent with previous evidence. Thomee [35] reported that excessive smartphone use was associated with disrupted sleep and insomnia, while Zencici et al. [36] found similar results among university students. Kumar et al. [21] and Meera et al. [22] also documented significant associations between smartphone addiction and poor sleep quality. In addition, Zhai et al. [37] identified links between prolonged screen exposure, psychological stress, and sleep problems, whereas Stanković et al. [38] reported significant associations between problematic smartphone use and sleep disorders.

Sleep disturbance may not only be an outcome of problematic smartphone use but also a contributing factor. Poor sleep quality can impair emotional regulation and increase vulnerability to anxiety and depressive symptoms, potentially leading to greater reliance on smartphones. This raises the possibility of a cyclical relationship in which sleep disruption and excessive smartphone use mutually sustain one another.

The proportion of explained variance (23%–30%) indicates that smartphone addiction is a meaningful, though not exclusive, correlate of anxiety, depressive symptoms, and sleep disturbance. These effect sizes are broadly consistent with findings reported across other cultural contexts and support the multifactorial nature of psychological distress in university populations.

These results may be interpreted through an integrative theoretical lens combining the Compensatory Model of Excessive Technology Use, cognitive–behavioral perspectives, and the Pre-Sleep Cognitive Arousal Model. Although conceptually distinct, these frameworks converge on several shared mechanisms, including emotional compensation, sustained cognitive engagement, fear of missing out (FOMO), and displacement of restorative behaviors such as sleep.

More specifically, the association between smartphone addiction and depressive symptoms may reflect compensatory use, whereby individuals engage excessively with smartphones to manage negative effects. Associations with anxiety may be more closely related to cognitive–behavioral mechanisms such as hypervigilance, persistent attentional engagement, and difficulty disengaging from online stimuli. The relationship with sleep disturbance is particularly compatible with the Pre-Sleep Cognitive Arousal Model, which emphasizes cognitive and physiological activation linked to nighttime smartphone use. Supporting this interpretation, Kang et al. [24] and Li et al. [25] reported that smartphone addiction significantly predicted sleep disorders among university students.

Despite the consistency of these associations, the findings require cautious interpretation. Because the study employed a cross-sectional correlational design, no causal or directional conclusions can be drawn. Psychological distress may increase reliance on smartphones, problematic smartphone use may contribute to distress, or both processes may occur simultaneously.

Several methodological limitations should also be considered. First, the study did not control for potential confounding variables such as gender, socioeconomic status, academic stress, or pre-existing mental health conditions. These factors may have influenced the magnitude of the observed associations. Second, all measures were based on self-report and collected at a single time point, increasing the possibility of common method variance and inflated correlations. Third, construct validity was not re-evaluated in the present sample through confirmatory factor analysis, which may limit measurement precision.

A further conceptual issue concerns the status of smartphone addiction as a construct. Although widely used in contemporary literature, it is not formally recognized as a distinct diagnostic category. Accordingly, the present findings should be interpreted as associations between excessive smartphone use and psychological distress rather than evidence relating to a clinically established disorder.

The cultural interpretation of the findings also warrants caution. Although the study was conducted in an Egyptian context, no direct cultural measures were included, and no cross-cultural comparisons were undertaken. References to cultural influences should therefore be regarded as theoretical considerations rather than empirically verified conclusions. Future studies should incorporate direct assessments of cultural variables and comparative designs to examine these issues more rigorously.

From an applied perspective, smartphone addiction may function as a potential marker of psychological risk. However, such implications remain preliminary because the present study provides no intervention-based or causal evidence. Longitudinal and experimental research is needed to determine whether reducing problematic smartphone use leads to measurable improvements in anxiety, depression, or sleep outcomes.

At the institutional level, the findings may offer an initial rationale for incorporating digital well-being into student support services. Any practical implementation should remain evidence-based and await stronger intervention research.

It is also possible that the observed associations reflect shared underlying vulnerabilities rather than direct effects of smartphone use itself. Traits such as impulsivity, chronic stress exposure, poor emotion regulation, or pre-existing mental health difficulties may contribute simultaneously to problematic smartphone use and psychological distress. In this sense, smartphone addiction may operate more accurately as an indicator of broader vulnerability than as a primary causal factor.

Finally, the generalizability of the findings is limited by the use of a convenience sample drawn from two universities and by the gender imbalance in the sample. Despite these limitations, the study provides useful preliminary evidence of significant associations that merit further investigation using longitudinal, multi-method, and more representative research designs.

Conclusion

The findings of the present study provide preliminary evidence linking smartphone use to psychological distress and sleep disturbances.

In conclusion, the present study offers strong evidence that smartphone addiction is significantly associated with anxiety, depression, and sleep disturbances among university students. By combining statistical results, theoretical frameworks, and cultural insights, the study contributes to the understanding of these associations within an Arab context, while acknowledging that cultural influences were not directly examined.

Despite these contributions, several limitations should be acknowledged. Cross-sectional and correlational designs limit the ability to draw causal inferences, while the reliance on self-report measures may introduce response biases and common method variance. In addition, the absence of control variables and the use of a convenience sample may restrict the generalizability and precision of the findings.

Future research is encouraged to employ longitudinal and experimental designs to better establish causal relationships and temporal dynamics. Incorporating multivariate analytical approaches that control for potential confounding variables would provide a more accurate estimation of the unique contribution of smartphone use. Furthermore, the inclusion of objective or behavioral measures, as well as the examination of potential mediating and moderating mechanisms (such as sleep quality, emotional regulation, and stress), would contribute to a more comprehensive understanding of the observed relationships.

Overall, the present study provides preliminary but meaningful evidence that underscores the importance of examining smartphone use within a broader psychological and behavioral framework.

Limitations

Several limitations should be considered when interpreting the findings of the present study.

First, the cross-sectional and correlational design precludes any causal or directional inferences. The observed relationships should therefore be understood as statistical associations rather than evidence of cause–and–effect links. Longitudinal and experimental research is needed to clarify temporal dynamics and causal pathways.

Second, the reliance on self-report measures collected at a single time point may introduce response biases, including social desirability and recall inaccuracies, as well as common method variance that could inflate observed associations. Future studies are encouraged to incorporate multi-method approaches, including behavioral or objective measures and multi-informant data, to enhance measurement validity.

Third, the use of convenience sampling from two universities, along with the observed gender imbalance and absence of statistical weighting, may limit the representativeness and generalizability of the findings. Recruiting more diverse and representative samples using probabilistic or stratified sampling techniques would strengthen external validity.

Fourth, the analytical approach relied on unadjusted regression models that did not account for potential confounding variables such as socioeconomic status, academic stress, and baseline mental health. As a result, the observed associations may not reflect the independent contribution of smartphone addiction. Future research should employ multivariate models to provide more precise estimates.

Fifth, although the instruments used in this study have demonstrated strong psychometric properties in prior research, confirmatory factor analysis (CFA) was not conducted in the current sample. This may limit confidence in the stability of the measurement structure across contexts. Future studies are encouraged to re-examine factorial validity and test measurement invariance.

Sixth, cultural influences were not directly measured or operationalized, despite the study being situated within an Egyptian context. Accordingly, cultural interpretations remain inferential. Future research should incorporate culturally sensitive measures and cross-cultural comparative designs to better understand sociocultural influences.

Seventh, the concept of smartphone addiction itself remains theoretically and diagnostically ambiguous, as it is not formally recognized as a distinct clinical disorder. This may blur the distinction between normative high use and clinically meaningful impairment. Future work is encouraged to adopt more refined conceptual frameworks and clearly differentiate between levels of problematic use.

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Authors' contributions

First Author **:** Contributed to the conceptualization of the study and formulation of the research problem; designed the research methodology; developed the study instruments; supervised data collection; conducted the statistical analyses; and drafted the initial version of the manuscript.**Second Author** **:** Contributed to the development of the theoretical framework and the review of the relevant literature; participated in the interpretation and discussion of the findings, considering prior studies and psychological theories; and provided scholarly review and critical revision of the initial manuscript.**Third Author** **:** Contributed to the overall academic supervision of the study; reviewed and refined the scientific and linguistic formulation of the manuscript; ensured compliance with scientific publication standards and APA style (7th edition); and supervised data collection.**Fourth Author**Contributed to the development of the theoretical framework and the review of the relevant literature.All authors confirm their substantive contributions to the study, their review and approval of the final version of the manuscript, and their agreement to submit it for publication.

Data availability
The data used in this study were generated and analyzed by the authors. data are available from the first author upon reasonable request, in accordance with ethical considerations and the protection of participants' privacy.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with internationally recognized ethical standards. Before data collection, participants were verbally informed about the study objectives, the voluntary nature of participation, and their right to withdraw at any time without consequences. Confidentiality and anonymity were strictly maintained, and no personally identifiable information was collected. The study received institutional approval from the (IRB) at the Faculty of Arts, Fayoum University (Approval No. 25.4.21).

Competing interests
The authors declare no competing interests.

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